MGT 207: Operations Management

Process Analysis & Capacity Management — The Car Wash Simulation Lab

PHASE 1 (15 min)
Quick Briefing
- AutoSpa P&L rules
- 4 Screen metrics

→

PHASE 2 (20m + 10m)
Round 1: "Cold Run"
- Run Challenge 1 ONLY
- **STOP** before Challenge 2

→

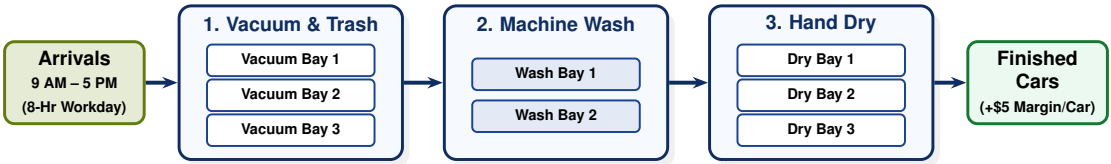
PHASE 3 (45 min)
Debrief Challenge 1
- Bottleneck & Sandwich Line
- Why variability breaks Ch. 1

→

PHASE 4 (60 min)
Graded Challenge 2
- Run Challenge 2 (Graded)
- Real-World & AI Playbook

"In operations, intuition gets you started—physics gets you paid. Today you will manage the exact same facility twice: once by gut feel, and once by operational science."

Your Role: Operations Manager at AutoSpa Car Wash



Rule 1: Spend 100% of $3,000,000 Budget

- Fixed **$3.0M budget** to upgrade station speeds.
- Must spend 100% every run:

$$(\text{Vacuum \$M}) + (\text{Wash \$M}) + (\text{Dry \$M}) = \$\mathbf{3.0}\text{M}$$

Rule 2: Maximize Daily Operating Profit

- Equipment depreciates over many years.
- $3M Capex **does NOT hit today's daily P&L**.
- Sole goal: maximize **Daily Operating Profit**.

How AutoSpa Makes Money & The Two Challenges

Daily Operating P&L (8-Hour Day)

- **Hours:** 9:00 AM – 5:00 PM (8 hours/day).
- **Price:** \$10 / car | **Variable Cost:** \$5 / car.
- **Profit per Car:** $\$10 - \$5 = \mathbf{\$5}$ / car.
- **Fixed Cost:** **\$675** / day *(break-even* $= 135$ *cars)*.

Challenge 1: Steady State (No Variation)

- **Constant Pace:** 37.5 cars/hr all day; fixed station speeds.
- **Gatekeeper:** Must hit max profit (\$825) to unlock Challenge 2!

Challenge 2: Real World (GRADED in Round 2!)

- **Variation:** Noon rush (**12–1 PM**) + random slowdowns.
- **STOP Rule:** Do **NOT** start Challenge 2 in Round 1!

FORMULA REMINDER: Daily Profit $= (\$5 \times$ Cars Washed$) - \$675$ | Cars Washed per Day $= 8$ Hours $\times$ (Cars Washed per Hour)

The 4 Numbers on Your Simulation Screen

1. Station Capacity (Speed)

- **What It Means:** Max cars a station can process per hour (**cars/hr**).
- **Why It Matters:** Investing Capex raises a station's hourly speed.

2. Capacity Utilization (%)

- **What It Means:** How busy a station is (0% to 100%).
- **Why It Matters:** 100% means running flat out; $<$ 100% means spare capacity.

3. Work-in-Process (WIP — Cars in Line)

- **What It Means:** Number of cars backed up waiting for a station.
- **Why It Matters:** A big line in front of a station tells you **that station is too slow**!

4. Throughput Time (Customer Wait Time)

- **What It Means:** Total **minutes** a customer spends at the car wash from arrival to exit.
- **Why It Matters:** More cars in line (WIP) = longer customer wait!

FORMULA REMINDER: Utilization (%) $= \frac{\text{Cars Processed per Hour}}{\text{Station Capacity}} \times 100\%$ | More Cars in Line (WIP) $\Longrightarrow$ Longer Customer Wait Time!

LAB SESSION 1 — Run Challenge 1 and STOP

RUN CHALLENGE 1 AND STOP!

Do NOT start Challenge 2 yet — Challenge 2 will be GRADED in Round 2!

Your Mission (Challenge 1 Only)

1. **Run Challenge 1 (No Variation):** Spend your \$3M across Vacuum, Wash, and Dry to hit **\$825 max profit**.
2. **STOP Once Solved:** Do **not** click Challenge 2!

Record 2 Things Before Break

1. Your winning **(Vacuum \$M, Wash \$M, Dry \$M)** split and number of trials needed.
2. Your winning **Cars Washed** (300/day) and **Wait Time** (19.0 min).

TIMING: 20-Min Challenge 1 Run → 10-Min Coffee Break → Reconvene for Debrief & Challenge 2!

Round 1 Debrief: Two Simple Rules of Any Process

Rule 1: The Bottleneck Governs Output

- A multi-stage line is only as fast as its **slowest stage**:

System Speed $=$ **Lowest Stage Speed**

- **On the Screen:** The bottleneck is the station with the **biggest line (WIP)** piled up right in front of it!

Rule 2: Why Lines Create Wait Time (Little's Law)

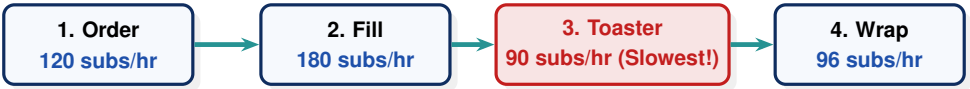
- Whenever cars arrive faster than a station can work, a **line (WIP)** builds up.

$$\textbf{Customer Wait Time} = \frac{\textbf{Cars in System (WIP)}}{\textbf{Process Speed}}$$

- **Takeaway:** The only way to cut customer wait time is to **keep lines (WIP) from piling up**!

FORMULA REMINDER: System Capacity = Lowest Stage Capacity (Slowest Stage) | Little's Law: More Cars in Line (WIP) $\Longrightarrow$ Longer Customer Wait Time

Warm-Up Case: The Sandwich Line (Shifting Bottleneck Trap)

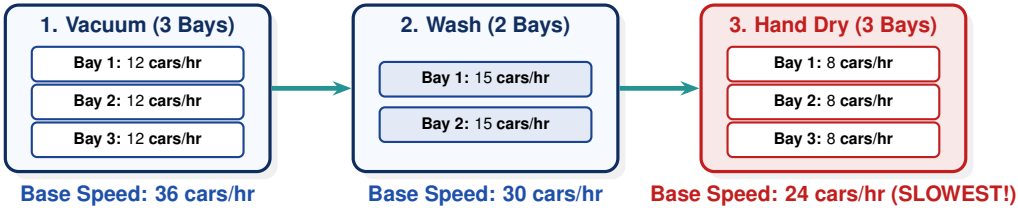


Quick Check (1 Minute — 2 Simple Questions)

1. **Starting Output:** With 1 Toaster (90 subs/hr), what is the sandwich shop's maximum hourly output?
2. **Buy a 2nd Toaster:** To fix the bottleneck, the manager buys a **2nd Toaster** (doubling Toaster capacity from 90 $\rightarrow$ **180** subs/hr). Does store output double to 180 subs/hr? What is the new output?

FORMULA REMINDER: System Capacity = Lowest Stage Capacity | 2 Toasters = 2×90/hr = 180 subs/hr

Debriefing Challenge 1: AutoSpa at $0 Spent



At $0 Spent, Hand Dry Caps the Entire Car Wash at 24 Cars/Hr:

- **Baseline Output ($0 Spent):** 8 hrs $\times$ 24 cars/hr $=$ **192** cars/day $\implies$ ($5 $\times$ 192) $-$ $675 $=$ $**285**/**day**.
- **Our Goal:** Spend our **$3.0M budget** to lift system speed from 24 cars/hr up to customer demand (**37.5** cars/hr)!

FORMULA REMINDER: Base Speeds ($0 Spent): Vacuum 36/hr, Wash 30/hr, Dry 24/hr | Customer Demand $=$ 37.5 cars/hr (300 cars/day)

Step 1: What Does $1 Million Buy at Each Stage?

Upgrade Price Tags (Shown at Bottom of Simulation Screen)

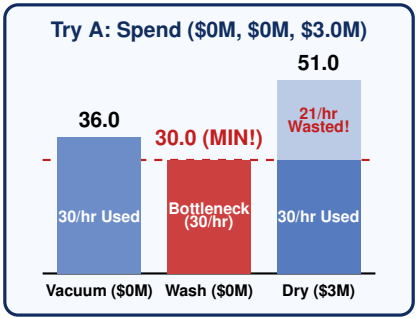
Each station starts at its **Base Speed ($0)** and gains a fixed number of cars/hr per **$1 Million spent**:

STAGE	BASE SPEED ($0)	GAIN PER $1M	NEW STAGE SPEED
1. Vacuum (3 Bays)	36.0 cars/hr	+**1.5** cars/hr	$36.0 + 1.5 \times$ ($M Spent)
2. Machine Wash (2 Bays)	30.0 cars/hr	+**15.0** cars/hr	$30.0 + 15.0 \times$ ($M Spent)
3. Hand Dry (3 Bays)	**24.0** cars/hr	+**9.0** cars/hr	$24.0 + 9.0 \times$ ($M Spent)

QUESTION: Hand Dry is slowest (24/hr). What happens if we dump **all $3M into Dry**, or split it evenly (**$1M** / **$1M** / **$1M**)?

FORMULA REMINDER: New Stage Speed $=$ Base Speed $+$ (Gain per $1M) $\times$ ($M Spent at Stage)

Step 2A: What If We Dump All $3M into the Bottleneck (Hand Dry)?



Naive Try A: ($0M Vacuum, $0M Wash, $3.0M Dry)

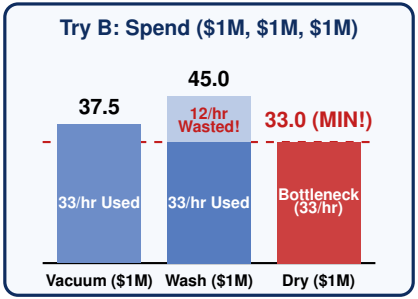
- **Vacuum ($0M):** 36.0 cars/hr
- **Wash ($0M): 30.0** cars/hr **(NEW BOTTLENECK!)**
- **Hand Dry ($3M):** $24.0 + 9.0(3) = \mathbf{51.0}$ cars/hr

Why It Fails (Just Like the Sandwich Toaster!):

- Bottleneck shifts to **Wash (**30 **cars/hr)**!
- **21.0** cars/hr of Dry capacity sits **100% wasted**!
- **Profit:** $8 \times 30 = 240$ cars $\implies$ **$525/day** *(loses $300/day).*

FORMULA REMINDER: Vac $= 36 + 1.5(\$M)$ | Wash $= 30 + 15(\$M)$ | Dry $= 24 + 9(\$M)$ | Profit $= (\$5 \times \text{Cars}) - \$675$

Step 2B: What If We Split the $3M Evenly ($1M / $1M / $1M)?



Naive Try B: ($1.0M Vacuum, $1.0M Wash, $1.0M Dry)

- **Vacuum ($1M):** $36.0 + 1.5(1) = \mathbf{37.5}$ cars/hr
- **Wash ($1M):** $30.0 + 15.0(1) = \mathbf{45.0}$ cars/hr *(12/hr wasted!)*
- **Hand Dry ($1M):** $24.0 + 9.0(1) = \mathbf{33.0}$ cars/hr **(BOTTLENECK!)**

Profit: $8 \times 33 = 264$ cars $\implies$ **$645/day** *(loses $180/day).*

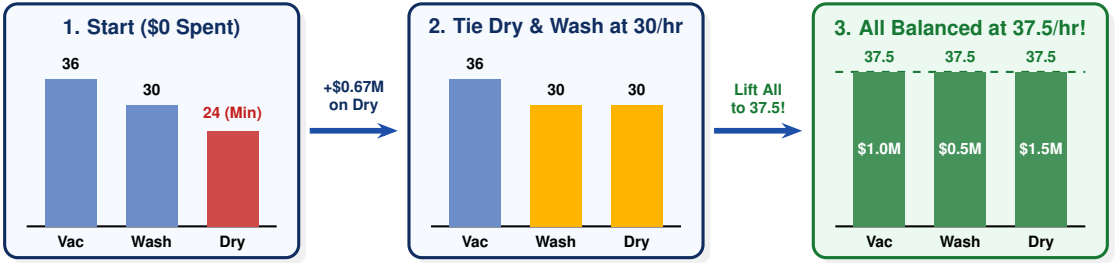
Golden Rule of Deterministic Capacity

Zero Capex is wasted ONLY when all 3 stages match:

Vacuum Speed = Wash Speed = Hand Dry Speed

FORMULA REMINDER: Vac $= 36 + 1.5(\$M)$ | Wash $= 30 + 15(\$M)$ | Dry $= 24 + 9(\$M)$ | Profit $= (\$5 \times \text{Cars}) - \$675$

Step 3: Spend Dollar by Dollar — Lifting the Lowest Bar!



Always Spend Your Next Dollar on the Lowest Bar:

1. **24 → 30/hr:** Spend \$0.67M on Dry $\implies$ **Dry & Wash tie at 30/hr**.
2. **30 → 36/hr:** Spend \$0.67M on Dry + \$0.40M on Wash $\implies$ **all 3 tie at 36/hr**!
3. **36 → 37.5/hr:** Remaining \$1.27M lifts all three together to **37.5** cars/hr!

FORMULA REMINDER: Vac $= 36 + 1.5(\$M)$ | Wash $= 30 + 15(\$M)$ | Dry $= 24 + 9(\$M)$ | Target Balanced Speed $= 37.5$ cars/hr

Step 4: The 3-Line Solution to Challenge 1 (Match All 3 to 37.5/hr)

Worked Example: Stage 1 (Vacuum)

- **Target Speed:** Customer demand is **37.5** cars/hr (300 cars/day).
- **Vacuum Equation:**

$$36.0 + 1.5 \times (\text{Vacuum \$M}) = 37.5$$

- **Solve for Vacuum $M:**

$$\text{Vacuum \$M} = \frac{37.5 - 36.0}{1.5} = \$\mathbf{1.00}\ \text{Million}$$

Your Turn (60 Seconds — 2 Quick Questions)

1. **Find Wash $M and Dry $M:** How much must you spend so Wash ($30 + 15 \times \$M = 37.5$) and Dry ($24 + 9 \times \$M = 37.5$) both hit 37.5 cars/hr?
2. **Find Max Daily Profit:** At 37.5 cars/hr (300 cars/day), what is your Daily Operating Profit?

FORMULA REMINDER: $\$M\ \text{Needed} = \frac{\text{Target Speed } (37.5) - \text{Base Speed}}{\text{Speed Gained per \$1M}}$ | $\text{Cars/Day} = 8 \times (\text{Cars/Hr})$ | $\text{Profit} = (\$5 \times \text{Cars/Day}) - \$675$

Preparing for Challenge 2: Why a “Perfectly Balanced” Line Fails

Rated vs. Usable Capacity

- **Rated Capacity:** Max speed under zero variability (Challenge 1).
- **Usable Capacity:** Realistic average speed with random slowdowns (Challenge 2).
- With variability:

$$\text{Actual } R < \text{Usable Cap.} < \text{Rated Cap.}$$

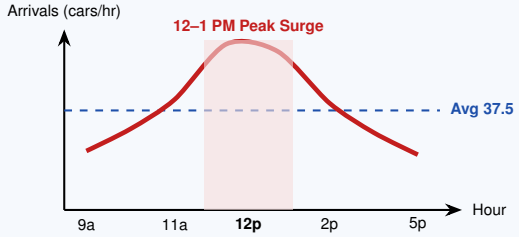
The 100% Utilization Trap

- Challenge 1 ran at **100% utilization** (37.5 cars/hr).
- In Challenge 2, **100% utilization causes gridlock**:
  - Slow hours immediately build a queue (WIP).
  - Zero spare sprint capacity to burn off the queue!

FORMULA REMINDER: Capacity Buffer = Rated Capacity − Average Demand | With Variability: Target Utilization (U) < 100%

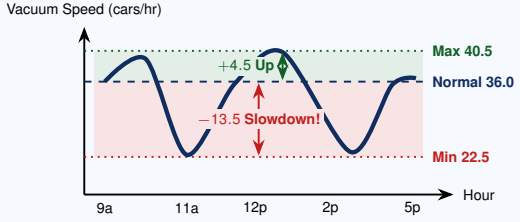
Anatomy of Variability in Challenge 2: Demand & Process

1. Demand Surge: 12–1 PM Rush



- Same $\sim$ 300 cars/day, but bunched at **11:30 AM – 1:30 PM**.

2. Station Speed Varies: Big Slowdowns



- Vacuum ($0M): Speedups $= +4.5$/hr, Slowdowns $= -\mathbf{13.5}$/hr!

FORMULA REMINDER: Vacuum ($0M): Normal $= 36.0$/hr | Best Hour $= 40.5$/hr ($+4.5$) | Worst Hour $= 22.5$/hr ($-\mathbf{13.5}$ Slowdown!)

How Variability Destroys Throughput: Starving & Moving Bottlenecks



1. Upstream Starvation

When Vacuum slows down, Wash and Dry sit idle. **Unused bay time at 10 AM cannot be saved for 12 PM!**

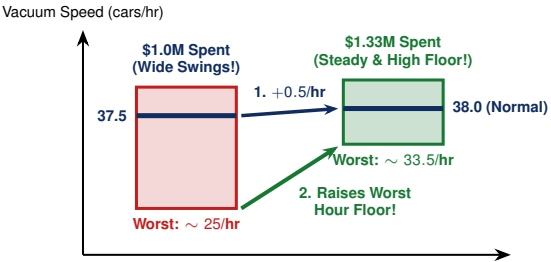
2. Midday Queue Spike

When Wash or Dry slows during the noon rush, WIP piles up and Wait Time (T) explodes.

3. Moving Bottleneck

Because station speeds fluctuate every hour, **the bottleneck wanders across stages**!

The Hidden Power of Capital in Challenge 2: Raising the Floor



Two Effects of Spending More on a Stage

1. **Raises Normal Speed:** Higher listed capacity.
2. **Shrinks Slowdowns (Raises the Floor):** Eliminates deep 22.5–25 cars/hr bad hours!

Why Challenge 1's Split Failed

Spending only $1.0M on Vacuum left Stage 1 exposed to deep hourly slowdowns—starving Wash and Dry!

Tactical Playbook for Graded Challenge 2

1. Protect Stage 1 (Anti-Starvation)

- **Vacuum is the gatekeeper:** Shift Capex **upstream into Vacuum (and Wash)** to eliminate slowdowns.

2. Don't Equalize Listed Speeds!

- Do *not* match all 3 bars! **Hand Dry** can run at a lower listed speed if Vacuum feeds it steadily.

3. Check the "Compare" Tab Charts

- Watch the **12–1 PM peak** on the Hourly Utilization & WIP charts to see which stage chokes.

4. Test in $50k–$100k Steps

- Adjust budget in clean increments toward **270** cars/day (**$675** profit).

LAB SESSION 2 — Graded Challenge 2 (25 Minutes)

Your Graded Mission: Challenge 2 (Variation)

Allocate your $3,000,000 budget in **Challenge 2 (GRADED)** using upstream floor protection and queue diagnostics.

Bronze Tier	Silver Tier	Gold Tier (HBP Optimum)
$\geq$ 255 **Cars/Day** ($\geq$ \$600 **Profit**)	$\geq$ 265 **Cars/Day** ($\geq$ \$650 **Profit**)	270 **Cars/Day** (\$675 **Profit**)
Beats Challenge 1's split.	Strong upstream protection.	Optimal variance buffering!

RECORD FOR GRADING: Best (Vacuum $M, Wash $M, Dry $M) split, Cars Washed, Daily Profit, and Wait Time!

FORMULA REMINDER: (Vacuum \$M) + (Wash \$M) + (Dry \$M) = \$3.00M | Daily Profit = (\$5 $\times$ Cars Washed) $-$ \$675

Final Debrief: Comparing Challenge 1 vs. Challenge 2 Strategies

1. Where Did Capex Move?

- Compare against Ch. 1's ($1.0M, $0.5M, $1.5M).
- Why did shifting money **from Dry to Vacuum/Wash** boost output?

2. "Flat Optimum" Effect

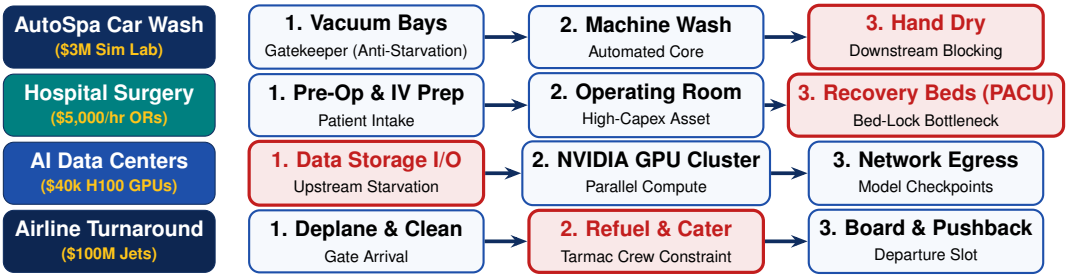
- Multiple nearby splits hit $\sim$ 270 cars ($675 profit).
- Moving bottlenecks create a **flat plateau** rather than a sharp peak.

3. Cost of Variability

- Same 300-car avg demand and $3M budget.
- Variability destroys **30 cars/day** and **$150/day** even under optimal management!

Core Lesson: Without variability, invest to **balance speeds**. With variability, invest to **raise upstream floors and buffer variance**.

Beyond the Car Wash: Same Physics, Billion-Dollar Stakes



Every Multi-Stage Operation Obeys the Exact Same Rules

- **Only the nouns change:** “Cars” become surgical patients, AI training batches, or Boeing 737s.
- **Starvation & blocking govern ROI:** A $5,000/hr Operating Room or $40,000 GPU is worthless if starved upstream or blocked downstream.

Healthcare Operations: Why $5,000/Hr Operating Rooms Sit Idle



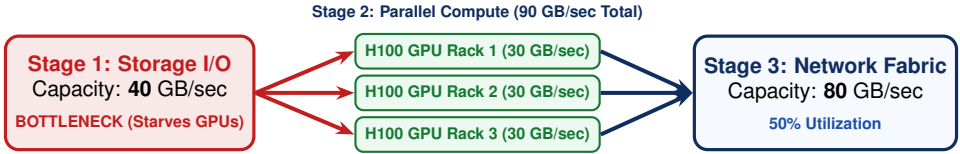
The "PACU Bed-Lock" Trap

- Just like AutoSpa's **Hand Dry**, when Stage 3 recovery beds fill up at 1 PM, surgeons cannot move patients out of the OR.
- The hospital's most expensive asset ($5,000/hr OR) becomes a very expensive waiting room.

The Operations Fix

- **Don't build more ORs:** Adding Stage 2 surgical suites yields **zero extra surgeries** while Stage 3 is the constraint.
- **Elevate Stage 3:** Discharge lounges and fast-track recovery protocols unlock existing OR capacity for 10% of the cost.

AI Infrastructure: Starving $40,000 NVIDIA GPUs



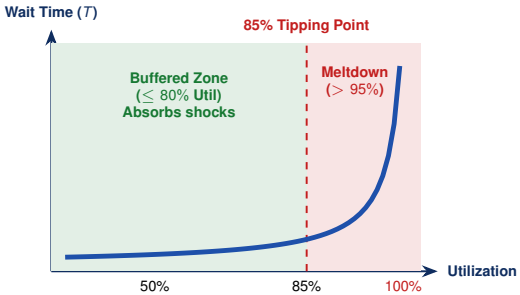
The "Subway Toaster" Mistake in AI

- Tech firms spend $100M doubling parallel NVIDIA H100 GPUs (Stage 2), expecting 2× faster LLM training.
- Instead, GPUs sit **55% idle** waiting for Stage 1 storage servers to feed training tokens (just like AutoSpa's Vacuum).

Protect Stage 1 to Feed the Core

- **Upstream floor protection:** High-speed NVMe caching and pre-fetching pipelines keep Stage 1 ahead of GPU demand.
- Always audit the **data ingestion pipeline** before buying another rack of GPUs.

Aviation & Fulfillment: Why 98% Utilization Guarantees Gridlock



The Kingman Hockey-Stick Curve

- Under variability, queue wait time explodes **exponentially** once utilization passes **85**%.
- Scheduling airlines or fulfillment centers at **98% utilization** leaves zero buffer to recover from a 15-minute weather delay.

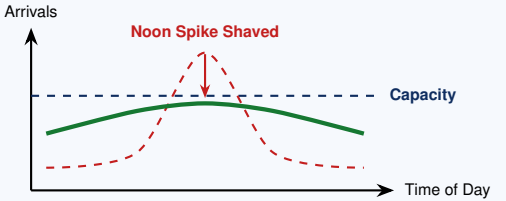
Strategic Capacity Buffers

- **Southwest & FedEx:** Intentionally hold **15–20% buffer capacity** (spare aircraft, slack sortation lanes) so one local hiccup never cascades into a multi-day network meltdown.

FORMULA REMINDER: Wait Time (T) $\propto$ Variability $\times \left(\frac{\text{Utilization}}{100\% - \text{Utilization}}\right)$ | At 100% Utilization + Variability $\Longrightarrow$ Infinite Queue

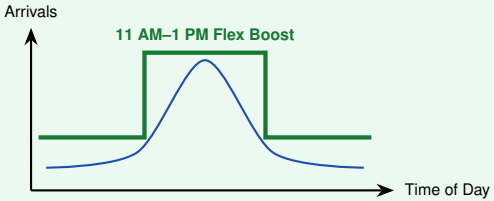
Surviving Peak Surges: Two Classic Executive Playbooks

Playbook 1: Reshape Demand (Shave the Peak)



- **Dynamic pricing:** Early-bird car wash discounts, Uber surge pricing, off-peak electric EV charging rates.
- **Appointment slotting:** Smooth random arrivals into an even stream below the capacity ceiling.

Playbook 2: Flex Capacity (Chase the Peak)



- **Cross-trained "swarm" labor:** Shift detailers or back-office staff to Hand Dry only during the 11 AM–1 PM rush.
- **Customer self-service:** Offload Stage 1 bottlenecks via mobile check-in kiosks and automated license-plate gates.

Enter AI: Unlocking “Virtual Capacity” Without Pouring Concrete

Traditional “Concrete” Capex

- Build more wash bays, ORs, or gates
- **Cost:** Millions in fixed Capex
- **Lead Time:** 12–36 months
- **Flaw:** Sits idle 70% of the day (off-peak)

VS

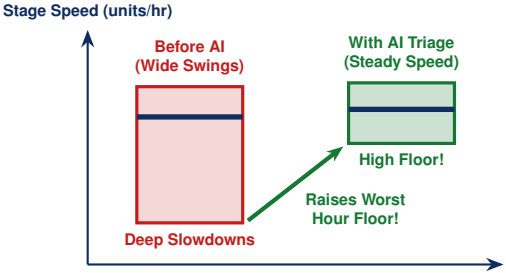
AI “Virtual Capacity” (Software)

- Attacks **variance**, **spikes**, and **routing**
- **Cost:** 5–10% of physical expansion
- **Lead Time:** Weeks to deploy
- **Payoff:** Unlocks 10–20% hidden throughput

How AI Rewrites the Capacity Equation

- Remember Challenge 2: **Variability stole 30 cars/day (10% of capacity)** even with optimal Capex.
- AI does not wash cars or perform surgery—it **reclaims the capacity lost to variability** across three operational levers.

AI Lever 1: Eliminating Slowdowns (Raising the Floor)



How AI Eliminates Worst-Case Slowdowns

- **Computer-vision intake triage:** Pre-scans arriving cars/patients to flag extreme outliers (*mud-caked SUVs, complex IVs*) and route them to a dedicated bypass lane before they block Stage 1.
- **Predictive maintenance:** Vibration and thermal ML models fix conveyor/pump micro-stalls overnight, cutting unplanned peak-hour downtime by **30–50**%.

FORMULA REMINDER: Effective Peak Capacity $\approx$ Determined by Minimum Floor Speed (Not Average Speed!)

AI Lever 2: Predictive Slotting (Taming the Noon Rush)



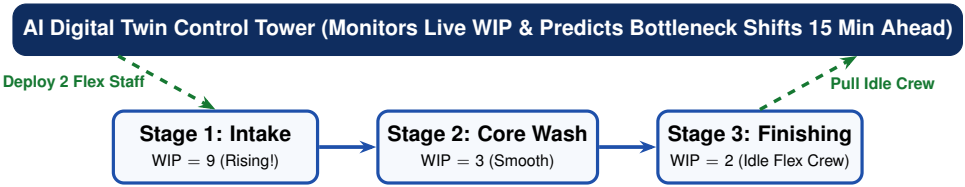
Why Standard Schedules Collapse

- Hospitals book ORs using **historical surgeon averages** (e.g., “hip replacement = 90 min”).
- When two complex cases land back-to-back at noon, afternoon queues explode and late surgeries are canceled.

Real-World Impact: UCHealth & Cleveland Clinic

- ML models predict exact procedure duration from patient EHR risk factors, surgeon, and anesthesia team.
- Interleaving high-variance and low-variance cases unlocked **10–15% more surgeries per room** with zero new concrete.

AI Lever 3: Real-Time Digital Twins & Dynamic Routing



Chasing the Moving Bottleneck in Real Time

- In Challenge 2, your Capex split was **locked for all 8 hours** even as the bottleneck shifted hour by hour.
- An AI Digital Twin tracks live WIP queues via cameras/sensors and **reallocates mobile capacity every 15 minutes**.

Industry Examples

- **Amazon Fulfillment:** Dynamically re-routes autonomous Kiva robots away from congested pack stations before gridlock forms.
- **Emergency Departments:** Triggers “swarm triage” alerts when Stage 1 waiting-room WIP crosses a 15-minute threshold.

C-Suite Playbook: When to Buy Concrete vs. When to Deploy AI

1. Steady Demand > Rated Capacity

- **Symptom:** All stages run at 95%+ utilization all day even with zero variance (like Challenge 1).
- **Prescription:** **Buy Physical Capex** — expand parallel bays at the bottleneck until balanced.

2. High Process Time Variance

- **Symptom:** Average capacity exceeds demand, yet Stage 1 slowdowns starve the line during peak hours.
- **Prescription:** **Deploy AI Lever 1** — computer-vision intake triage + predictive maintenance to raise the floor.

3. Predictable Peak Arrival Surges

- **Symptom:** Facility is half-empty at 9 AM but gridlocked from 11 AM to 2 PM.
- **Prescription:** **Deploy AI Lever 2** — ML duration slotting + dynamic off-peak pricing to flatten the peak.

4. Shifting Hourly Bottlenecks

- **Symptom:** Queues jump unpredictably between Upstream Intake and Downstream Finishing.
- **Prescription:** **Deploy AI Lever 3** — Digital Twin real-time crew swarming + cross-trained labor buffers.

FORMULA REMINDER: **Golden Rule:** Never approve $1M in physical concrete until you have squeezed out the variability with software and process flow!

Executive Mini-Case: The Noon Backlog at MetroHealth Imaging



The Situation on Your Desk (8-Hour Operating Day)

- **Demand vs. Rated Capacity:** Daily demand is **32** patients/day (4.0 pts/hr). Your 2 MRI scanners can handle **40** patients/day (5.0 pts/hr).
- **The Problem:** Half of all patients book between **11 AM and 2 PM**, and difficult Stage 1 IV insertions drop Stage 1's floor to **2.0** pts/hr—starving the MRIs so the clinic only completes **28** scans/day (losing 4 scans/day at $1,000 margin/scan = $1M/yr).

FORMULA REMINDER: Daily Demand = 32 scans | Stage 2 MRI Rated Capacity = 40 scans | Actual Output = 28 scans (4 lost/day to starvation + peak surge)

You Make the Call: Concrete vs. Algorithms at MetroHealth

The Clinic Director's $3.0M Proposal

- **The Pitch:** "Patients wait 90 minutes at noon and we turn away 4 patients/day. We must spend $**3.0** Million to buy a **3rd MRI Scanner** (raising Stage 2 to 60 scans/day)."
- **Your Role:** As COO, you must evaluate this Capex request using Session 6 principles.

Your Turn (2 Executive Questions)

1. **Diagnose the $3.0M MRI Request:** Using your AutoSpa Challenge 2 lessons, why will adding a 3rd MRI scanner at Stage 2 fail to solve the noon backlog?
2. **Design a $< \$200$k Operational + AI Fix:** What two targeted interventions at **Stage 1** and **Appointment Scheduling** will capture all 32 scans/day without buying a 3rd MRI?

FORMULA REMINDER: System Output = Governed by Upstream Starvation & Peak Variability | 4 Extra Scans/Day $\times$ \$1,000 = $+$**\$4,000/day** (+\$1.0M/year)

Executive Summary: 4 Enduring Rules of Capacity Management

1. The Bottleneck Governs the P&L

- System capacity = **Lowest Stage Capacity**.
- Upgrading a bottleneck helps **only until the constraint shifts**; extra Capex beyond that is wasted.

2. Lines (WIP) Directly Drive Customer Wait Time

- By Little's Law, more cars backed up in line (WIP) directly inflates customer wait time.
- To cut customer wait time, you must prevent queues from forming at slow stations.

3. Service Capacity Is Perishable

- Idle bay time at 10 AM cannot be stored for 12 PM.
- **Never starve downstream stages:** protect Stage 1 reliability.

4. Variability Requires Capacity Buffers

- Targeting **100% utilization** under variability guarantees gridlock.
- Invest to **shrink variance and raise the minimum floor**.